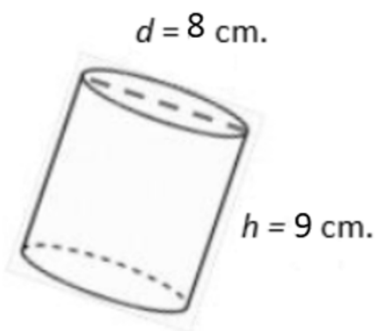
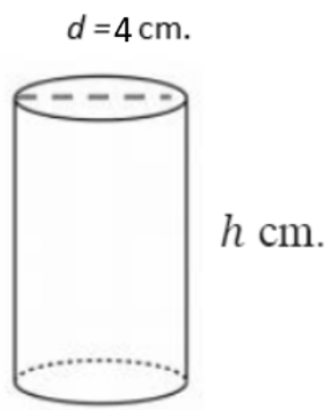
Grade 7
Unit 12
Lesson 6 Practice

Name Answer Key
Date
Period

Cylinder C and Cylinder D have the same volume in cubic centimeters.

Cylinder C	Cylinder D
	

1. Find the volume of Cylinder C. Leave it in terms of π.
$V = \pi r^2 h$
$V = \pi \cdot 4^2 \cdot 9$
$V = 144\pi \text{ cm}^3$

2. Write an expression for the volume of Cylinder D.

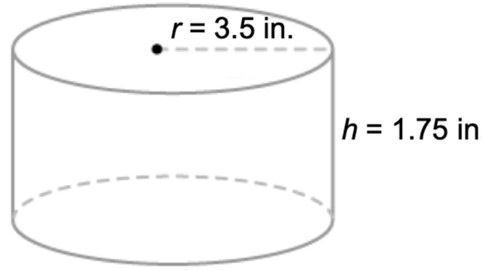
$V =$ $\pi \cdot 2^2 \cdot h$

3. Write an equation to show the volumes are equal to each other.
$\pi \cdot 2^2 \cdot h = 144\pi$

4. Solve the equation.
$\pi \cdot 2^2 \cdot h = 144\pi$
$\frac{\pi \cdot 4 \cdot h}{4\pi} = \frac{144\pi}{4\pi}$
$h = 36$ cm

5. What does the solution to the equation represent?
The height of cylinder D

6. Find the volume of the cylinder shown. Write your answer in terms of π.



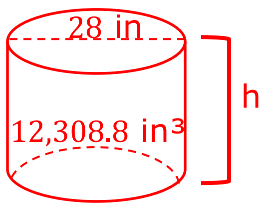
$V = \pi r^2 h$

$V = \pi \cdot 3.5^2 \cdot 1.75$

$V = 21.4375\pi \text{ in}^3$

A bucket can hold $12{,}308.8$ cubic inches. The diameter of the bucket is 28 inches (in.). What is the height of the bucket? Use 3.14 for π.

7. Sketch a diagram of the bucket and label the dimensions.



8. Find the height of the bucket.

$V = \pi r^2 h$

$12308.8 = 3.14 \cdot 14^2 \cdot h$

$12308.8 = 615.44h$

$h = 20$ in.

9. The figure shows a coffee cup. The diameter measures 2 inches (in.) and the coffee cup can hold up to 16.49 cubic in. What is the height of the coffee cup? Use 3.14 for π.

Diameter: 2 in

$V = \pi r^2 h$

$16.49 = 3.14 \cdot 1^2 \cdot h$

$16.49 = 3.14h$

$5.25\ in. \approx h$

10. A bucket in the shape of a cylinder is used to carry grain. The bucket has a diameter of 18 inches (in.) and can carry up to $6{,}104$ cubic inches of grain. What is the height of the bucket? Use $\frac{22}{7}$ for π and round to the nearest hundredth.

$V = \pi r^2 h$

$6{,}104 = \frac{22}{7} \cdot 9^2 \cdot h$

$6{,}104 = \frac{22}{7} \cdot 81 \cdot h$

$6{,}104 = \frac{1782}{7} h$

$h = 23.98$ in.